

AUDIT SURFACE FOR

Capability Matters: A Casebook

Validation & Audit

INDEXES · PER-CASE REFERENCES

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Learning Engineering for Next-Generation Systems · v2.1

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SECTION

What this document is for

Validation and Audit is the auditor-facing companion to **Capability Matters: A Casebook**. It carries the surfaces an instructor, reviewer, or accreditation reader needs to verify the casebook — the indexes that let any case be located by domain or by LENS course, and the per-case reference appendix that lets any source be traced. It is the same data the casebook itself ships, lifted into a standalone document so the LENS Companion can stay focused on what LENS **is** (the five competencies, the LEOs, the course mapping) without the audit pages travelling with it.

The document is in three parts:

- I. Cases by primary domain — every case classified by the operational domain it primarily evidences, so a domain expert can find the cases that touch their territory without scanning the table of contents.
- II. Cases by LENS course — every case classified by the LENS course it serves as a worked example for, so syllabus authors and capstone advisors can find the cases that fit a course's stated learning outcomes.
- III. References, case by case — every reference cited in every case, organised by case in number order, with an explicit **Retrieved from:** line per entry so a reviewer can locate the canonical source.

On verification. The running per-case verification track lives in the repository at `casebook/verification-log.md`, a 194-row table whose six auto-prefilled columns track the structural and sourcing checks the build pipeline can answer on its own. The seventh column — **conclusions reasonable** — is the human case-by-case content read this document supports. The references appendix below is the per-case checklist for that read; the **Retrieved from:** lines are the canonical-access record each case acquires as the verification pass closes.

DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganizes the cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Where a section opens with a callout, it names the dataset’s standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE

Case 128

Operation Eagle Claw (1980). Cross-organizational capability that existed on no service’s deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater-Nichols.

STANDOUT SUCCESS

Case 137

U.S. Nuclear Navy / Rickover Training Model (1954–). The longest continuous capability-engineering program in any high-consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.

124	USS Fitzgerald & USS John S. McCain	2017	T-K-N
126	F-35 Sustainment & the Maintainer Capability Gap	ongoing	T-K-D
127	Military Fratricide — Desert Storm to Afghanistan	1991 – 2014	T-H-K
128	Operation Eagle Claw	1980	T-K
129	Patriot Missile / Dhahran	1991	D-H-K
130	USS Vincennes & Iran Air Flight 655	1988	H-T
131	Stanislav Petrov / 1983 False Alert	1983	H-T
133	Mark 14 Torpedo Failures	1941 – 1943	T-K-G
134	V-22 Osprey	1991 – present	T-H-N
135	9/11 Intelligence Sharing Failures	1996 – 2001	G-K
137	U.S. Nuclear Navy / Rickover Training Model	1954 – present	T-K-N
138	MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard	2020 revision (series since 1968)	G-K

139	GIFT and the Adoption Gap	2012 – present	K·G·N
140	Navy Surface Warfare Readiness Reform	2018 – present	T·K·N
141	DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks	2009 – 2014	H·K·D

AVIATION

STANDOUT FAILURE

Case 97

Boeing 737 MAX / MCAS (2018–19). A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS

Case 117

Crew Resource Management & CAST (1981–). Authority gradients named and engineered into the cockpit, then a closed-loop industry data architecture sixteen years later. The 83% fatality-risk reduction CAST reports is a portfolio result; CRM's separate contribution is not isolated in any published estimate.

96	AeroPerú Flight 603	1996	H·T
97	Boeing 737 MAX / MCAS	2018 – 2019	D·T·H
102	Air France Flight 447	2009	T·H
103	Kegworth / British Midland 92	1989	T·H·K
104	Asiana Airlines Flight 214	2013	T·H
105	Helios Airways Flight 522	2005	T·H
106	TransAsia Airways Flight 235	2015	T·H
107	Eastern Air Lines Flight 401	1972	H·T
109	Colgan Air Flight 3407	2009	T
110	Atlas Air Flight 3591	2019	T·K
112	NASA Saturn V Documentation	1972 – present	K
117	Crew Resource Management & CAST	1981 – present	T·H·N
118	Korean Air Safety Transformation	2000 – present	T·N
119	Aviation Safety Reporting System (ASRS)	1976 – present	T·K·N
120	EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation	1996 – 2002	H·K·G
121	TCAS — Coordinated Collision Avoidance and the Überlingen Lesson Version 7.1)	1981 – 2008 (TCAS II)	H·K·G
122	Singapore Airlines Safety Transformation	1980s – present	T·N

132 F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap 2008 – 2012 **H-K-N**

185 Air Canada Chatbot Liability — Delegation Without Revocation 2022 – 2024 **D-K-N**

HEALTHCARE

STANDOUT FAILURE

Case 2

EHR / CPOE Implementation (2005–). Roughly \$30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.

STANDOUT SUCCESS

Case 23

WHO Surgical Safety Checklist (2008–). One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

1 Therac-25 1985 – 1987 **H-D-G**

2 EHR / CPOE Implementation 2005 – present **H-D-G**

4 Mid Staffordshire NHS Foundation Trust 2005 – 2009 **G-N-K**

5 Epic Sepsis Model 2017 – 2021 **D-G-N**

8 Medical Errors as Systemic Failure 1999 – present **T-H-N-K-G**

9 Vioxx Withdrawal 1999 – 2004 **G-D**

11 California Nurse Staffing Ratios — Legislating a Capability Requirement 2004 – 2010 **G-K**

12 ACGME 80-Hour Resident Duty-Hour Reform 2003–2017 **T-K-N**

13 Implementation Science in Healthcare — The 17-Year Gap ongoing **K-G-N**

19 Keystone ICU / Pronovost Checklist 2004 – present **T-N**

20 TREWS / Sepsis Watch 2018 – 2022 **H-K-G**

21 Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff 2024 – 2025 **H-K-N**

22 ChatGPT in Healthcare — Hallucination Cases 2023 – present **H-D**

23 WHO Surgical Safety Checklist 2008 – present **T-N**

24 Bristol Heart Babies Reform 1984 – present **G-N**

32 Annual-Screening UI Redesign + CDS at University of Missouri Health Care 2020 **T-D-N**

33 Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal 2019 – 2024 **T-D-N**

34 COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case 2022 – 2024 **T-K-D**

35 Radiology AI Miscalibration 2018 – present **H-K-D**

36	AlphaFold — Protein Structure Prediction	2020 – present	T
37	DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback	2020	T·K·D
39	TeamSTEPPS	2006 – present	T·N
176	Tylenol Recall	1982	G·N

EDUCATION AT SCALE

STANDOUT FAILURE

Case 54

Summit Learning / Personalized Learning Rollout (2014–19). A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.

STANDOUT SUCCESS

Case 80

Georgia State University Predictive Analytics (2012–). 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

45	Tennessee Voluntary Pre-K Study	2009–2018	G·D
46	Algorithmic Bias in Educational Predictive Analytics	ongoing	G·H·D
51	Atlanta Public Schools Cheating Scandal	2009 – 2015	G·N
53	inBloom	2014	G
54	Summit Learning / Personalized Learning Rollout	2014–2019	G·T·K
67	Cognitive Tutor / Carnegie Learning	1990s – present	T
80	Georgia State University Predictive Analytics	2012 – present	T·K
142	xAPI / Total Learning Architecture — Interoperability Gap	ongoing	K·G
205	The Discipline We Build Next	ongoing	T·K·N·G·H·D

ENERGY

STANDOUT FAILURE

Case 166

Three Mile Island (1979). Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.

STANDOUT SUCCESS

Case 175

INPO and the Nuclear Academy (1979–). The industry built the institution TMI revealed it needed. No INES Level 4-or-above accident at a U.S. commercial reactor since.

146	Northeast Blackout	2003	H·K
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160	Davis–Besse Reactor Head Corrosion	2002	N·K·G
161	Texas City BP Refinery Explosion	2005	N·T·K·G
162	Upper Big Branch Mine Explosion	2010	N·G·K
163	Deepwater Horizon	2010	T·N·K
165	Camp Fire / PG&E	2018	G·N·K
166	Three Mile Island	1979	T·H
169	Fukushima Daiichi	2011	N·G·K
175	INPO and the Nuclear Academy	1979 – present	T·K·G

GOVERNMENT

STANDOUT FAILURE

Case 191

Australia Robodebt (2016–2023). An income-averaging algorithm that reversed the burden of proof onto welfare recipients, raising some 470,000 wrongful debts. The Royal Commission found the scheme unlawful and attributed deaths to its operation — circumstantially, not as a finding of individual causation.

NOTE

Case 203

The reforms that follow these failures — Royal Commissions, statutory restructures — are documented inside the narratives rather than as separate cases. The government dataset’s paired-intervention success is NYC Local Law 144 (Case 203), filed under its own primary domain.

7	VA Wait–Time Scandal	2014	G·K·N
180	Healthcare.gov Launch	2013	K·T·G
193	Bernard Madoff / SEC Failures	1992 – 2008	G·K·N
197	Predictive Policing — PredPol	2011 – present	G·H·D

INDUSTRIAL

STANDOUT FAILURE

Case 159

Bhopal (1984). Training, knowledge, normalization of deviance, and governance all collapsed in the same plant. Thousands died within hours; total-death estimates run to 15,000–20,000, with roughly half a million exposed. The cautionary case for outsourced operational responsibility.

STANDOUT SUCCESS

Case 155

Toyota Production System / Andon Cord (1950s–). Any operator can signal a defect at the source. Under the fixed-position stop system most pulls never halt the line — the smallest authority-architecture intervention in the dataset, and the most durably copied.

144	Kodak Digital Camera Stagnation	1975–2012	D·K·N
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147	Takata Airbag Inflators	2008 – 2023	D-G
149	Volkswagen Dieselgate	2015	D-G
151	GM Ignition Switch	2002 – 2014	D-G
155	Toyota Production System / Andon Cord	1950s – present	N-G
159	Bhopal	1984	T-K-N-G
164	Grenfell Tower	2017	G-T-K-N

TECHNOLOGY

STANDOUT FAILURE	Case 184	UK Post Office Horizon Scandal (1999–2015). Institutional refusal to believe sub-postmasters’ reports of computer error, sustained over fifteen years. More than 900 wrongful convictions, 236 imprisoned, and 13 suicides the inquiry could not rule out as Horizon-caused — the longest-running operator-feedback failure in the dataset.
NOTE	Case 36	The technology dataset’s success story is filed under Healthcare: AlphaFold (Case 36), the contemporary worked example of computational capability-engineering done as a discipline.

143	Knight Capital Trading Loss	2012	D-K
148	Wells Fargo Fake Accounts	2011 – 2016	G-N
152	TSB Bank IT Migration	2018	H-G
153	Equifax Data Breach	2017	G-K
157	AI-Augmented Coding Tools	2021 – present	T-H
172	CrowdStrike Falcon Outage	2024	D-K-G
184	UK Post Office Horizon Scandal	1999 – 2015	G-H-K

AI IN EDUCATION

88	LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models	2025	T-K-N
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K-12 EDUCATION

48	School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism	2010s – 2022	G-K-N
60	Houston EVAAS — Value-Added Teacher Evaluation on Trial	2011–2017	G
61	Sold a Story — The Science of Reading and the Decades-Long Research-Practice Gap	1997–2023	K-T
62	Gates Intensive Partnerships — Measurement Without a Coupled Lever	2009–2018	G-K

63	LAUSD iPad Program — The Common Core Technology Project	2013–2015	D·H·G
66	Growth–Mindset National Experiment — Heterogeneous Effects	2019	D·N·K
75	Chen et al. — Rural China AI Devices and the Equity–Direction Finding	2025	T·K·D
95	PBIS Implementation Fidelity — Why the Framework Travels Only with Its Measurement Loop	1997–2024	T·G

K–12 MATHEMATICS

72	ASSISTments — National Replication and Long–Term Follow–Through	2014 – present	T·K·D
84	Cognitive Tutor Algebra I at Scale — Year–One Null, Year–Two Positive	2007 – 2014	T·K·D

K–12 SCIENCE

74	Zhang/Scardamalia — Knowledge Building Across Three Cohorts	2009	T·K·D
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AIR TRAFFIC MANAGEMENT

116	Eurocat ATM Pilot Modernization — Small–Tier Verification as the Gateway to Big–Tier Change	2005	G·K·H
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ANESTHESIOLOGY

27	Anesthesia Monitoring Standards and the APSF — The Mortality Decline	1986 – present	H·K·G
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AUTONOMOUS SYSTEMS

STANDOUT FAILURE

Case 183

Uber ATG / Tempe Fatality (2018). The safety driver was nominally responsible for what the system was designed to do without her attention — the classic vigilance contradiction at the heart of partial-autonomy operator roles.

NOTE

Case 205

This key’s cases are all failures and frontiers; the paired-intervention successes — Waymo’s safety case (199) and the CPUC permit regime (200) — are filed under autonomous vehicles. See Case 205 (**The Discipline We Build Next**) for the open question this volume closes on.

183	Uber ATG / Tempe Fatality	2018	T·N·G·H
195	Tesla Autopilot — Recurring Fatalities	2016 – present	T·N·G·H

AUTONOMOUS VEHICLES

190	Cruise’s Partial Disclosure — How Disclosure Posture Decides Deployment	2023 – 2024	G·K·N
199	Waymo’s Safety Case Framework — Governance Objection Dissolved by Designed Artifact	2023 – 2026	G·K·N

200	CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver 2020 – 2026	G·K·D
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AVIATION INFRASTRUCTURE

115	FAA NextGen and the ADS-B Out Transition 2003 – 2020	G·D·K
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AVIATION SAFETY

114	FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule 1988 – 2010s	G·D·K
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123	FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike 1992 – 2000s	G·K·N
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CLINICAL CARE

15	I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer 2014	H·K·N
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29	Bar-Code Medication Administration — Cue at the Point of Care 2010	H·K·D
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CLINICAL MEDICINE

6	Racial Bias in Pain Assessment — The False-Belief Mechanism 2016	T·K·N
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25	Removing the Race Coefficient from eGFR 2021	D·G·N
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CLINICAL/TRANSLATIONAL RESEARCH

40	Team Science Training for Clinical and Translational Scientists 2020 – 2022	K·N
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CORPORATE L&D

65	Training Transfer — The Gap Between Learning and Doing 2010	K·N
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70	High-Impact Learning System — Engineering the Environment, Not Just the Event 2001 – present	K·N
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79	The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops 1959 – present	K·N
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83	Brinkerhoff Success Case Method — Tails as the Evaluation Instrument 2005 – present	K·N
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CRIMINAL JUSTICE

187	COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate 2014 – 2018	D·K·N
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DIGITAL IDENTITY

201	Aadhaar Exclusion — Biometric Welfare Delegation and Its Judicial Reckoning in India 2018 – 2025	G·N·H
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E-GOVERNMENT

194	Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox) 2001 – present	G·K·N
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ED-TECH

47 Algorithmic Bias in Automated Exam Proctoring 2022 **D·N·K**

EDUCATION (NORWAY)

92 Norway's National Expert Commission on Learning Analytics 2021 – 2023 **G·K·N**

EDUCATION AT SCALE

50 Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction 2012 – 2023 **D·K·N**

69 Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale 2016 **T·D**

EMERGENCY MANAGEMENT

167 Hurricane Katrina — The FEMA/DHS Response Failure 2005 **G·K**

171 Hurricane Maria — Puerto Rico Response and Logistics Failure 2017 – 2018 **G·N**

FINANCE

186 Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity 2009 – 2022 **D·G·N**

196 Fintech Lending Fairness Audit — When Including Race Reduces Disparity 2025 **D·G·N**

FINANCIAL SERVICES

192 Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model 2019 – 2021 **D·K·N**

GENERAL AVIATION

100 Glass-Cockpit Transition in Light Aircraft — Technology Outran Training 2010 **H·N·K**

GLOBAL HEALTH

18 PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption 2023 **K·N·D**

43 Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface 2013 – 2018 **H·N**

170 West Africa Ebola — The Delayed International Response 2014 – 2016 **G·N**

GOVERNMENT

49 UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days 2020 **D·K·N**

191 Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict 2016 – 2023 **D·G·N**

203 NYC Local Law 144 — Bias Audits as Governance Artifact 2023 – present **D·G·K**

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189 Dutch SyRI — Welfare–Fraud Risk Scoring Halted on Rights Grounds 2014 – 2020 **D·G·N**

HEALTH PROFESSIONS EDUCATION

28 Interprofessional Education and the Evidence Gap 2013 – 2015 **K·N**

HEALTHCARE/ONCOLOGY

3 Watson for Oncology — Delegation Marketed Ahead of Capability 2013 – 2018 **D·K·N**

HIGHER EDUCATION

55 Enrollment–Algorithm Yield Optimization Across U.S. Higher Education 2010s – present **T·K·N**
(Brookings synthesis 2021)

57 GAO Online Program Manager Oversight Gap (GAO-22-104463) 2022 **D·K·N**

58 USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC) 2010s – 2024 **D·K·N**

85 OU Analyse — Predictive Learning Analytics and Teacher Use at Scale 2019 – 2023 **T·K·D**

86 Gándara — Detecting and Mitigating Algorithmic Bias in College Student–Success Prediction 2024 **D·K·N**

90 Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions 2025 **T·K·N**

HIGHER EDUCATION (AFRICA)

89 Data Privacy and Learning Analytics on the African Continent 2022 **K·G·N**

HIGHER EDUCATION (BRAZIL)

82 MMALA — A Maturity Model for Responsible Learning–Analytics Adoption (Brazil) 2024 **K·N**

HIGHER EDUCATION (LATIN AMERICA)

91 LALA — Building Learning–Analytics Governance Capacity Across Latin America 2017 – 2020 **K·N**

HIGHER ED (UK)

81 Open University 'Ethical Use of Student Data' and OU Analyse 2014 – 2025 **G·K·N**

HIGHER-ED ANALYTICS

52 Purdue Course Signals — The Reverse–Causality Retention Claim 2012 – 2013 **D·K·N**

HOSPITAL PHARMACY

42 Australian Hospital–Pharmacy Technician Role Redesign 2016 **D·N·H**

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- 87** Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models 2021 – 2024 **D·K·N**
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- 94** Learning Analytics on the African Continent — A Scoping Review as the Present-State Map 2022 **K·N**
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- 26** Pulse Oximetry Across Skin Tones 1990 – 2020 – 2025 **D·G·N**
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- 17** Spaced Education RCTs in Medical Training 2007 – 2009 **H·K·D**
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- 76** Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account 2023 **T·K·N**
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- 198** Launching the BRAIN Initiative — Governance of a Grand Challenge 2011 – 2015 – present **G·K·N**
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- 78** CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale 2017 – present **H·K·N**
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156	INL / LWRs Control-Room Modernization — Sustainment Research for an Aging Fleet 2010 – present	D·H·K
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154	INL Turbine-Control Upgrade — Verification and Validation as the Cutover Deliverable 2014	G·K·H
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NURSING EDUCATION

16	NCSBN National Simulation Study — Licensing the 50% Substitution Rule 2014	G·K·D
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PHARMA SAFETY

38	iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms 2006 – 2011	G·D·N
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PUBLIC HEALTH

179	The Johns Hopkins COVID-19 Dashboard — Evidence Infrastructure at Decision Speed 2020 – 2023	G·K
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RAIL TRANSPORT

177	CIRAS — Confidential Incident Reporting for UK Rail 1996 – present	G·K·N
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SOFTWARE ENGINEERING EDUCATION

71	Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development 2025 (preprint)	K·N
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SOFTWARE SUSTAINMENT

174	Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed 1996 – 2000	G·D·K
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SPACE / AEROSPACE

STANDOUT FAILURE	Case 111	Challenger & Columbia (1986 / 2003). Normalisation of deviance across seventeen years and two crews — the sociologist Diane Vaughan’s term for a culture that accepted flying with flaws once problems were defined as normal and routine, and the culture the CAIB found still intact in 2003.
NOTE	Case 137	No paired-intervention success in the space dataset — Part III’s aviation half carries seven. See Defense (Rickover, Case 137) for the safety-engineering counterpart human spaceflight did not build.

98	Mars Climate Orbiter — Unit Mismatch 1999	D·K
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101	Ariane 5 Flight 501 1996	D·K·H
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108	NASA EVA-23 — Water Intrusion Inside a Spacesuit	2013	H·N·D
111	Challenger & Columbia	1986 / 2003	N·K·G
113	Boeing Starliner	2019 – 2025	K·D

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30	Surgical Skill Video Peer-Rating Predicts Complications	2013	H·D·K
31	Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units	2006 – 2016	T·K·H

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182	Amazon Hiring AI — Trained Bias, Deprecated 2017	2014 – 2018	D·K·N
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77	Hybrid Human-AI Tutoring — Augmentation, Not Delegation	2024	T·K·D
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68	CIRCUIT — A Scalable, Equity-Centered Research Workforce Model	2017 – 2023 (six cycles)	T·K
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About this index. Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

LENS COURSE INDEX

The cases by LENS course.

Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the concentration, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

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36 cases

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Every reference cited in every case, organized by case in number order. The intent is that any reader can locate every source the book draws on. Each entry preserves the in-case citation form (author, year, venue) and adds a **Retrieved from:** rule beneath it — a worksheet line for the reviewer to enter the canonical access path they used: a publisher URL, a DOI, an agency landing page, or a stable archival locator. The rules print blank by design, because the per-case verification is being carried out against this appendix rather than reported by it; the running status of that work is in casebook/verification-log.md. Future editions will print the locators as the review closes them out.

Style. References are rendered in the form they appeared in the case — APA-style author/year/title/venue, with italicised book titles and journal names — supplemented by the explicit retrieval locator. Government and agency reports are cited by report number when one exists. Trade-book references that carry no DOI use ISBN as the stable identifier; that mapping is on the verification log.

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2020s

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2016

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2014 – present

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2009–2018

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ongoing

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